Software Requirements Specification (SRS)

Lane Management System

Authors: Graham Parker (section 5), James Chen (section 4), Sean Finkel (section 3), Mohsin Hossain (section 1&2), Shane Carr

Customer: Ayush Agrawal

Instructor: Betty H.C. Cheng

1. Introduction

This Software Requirements Specification (SRS) document will give a comprehensive explanation of the Lane Management System (LMS) that assists drivers to be in their lanes safely. The SRS provides the purpose of the system, individuals using the system, and what the system can and cannot accomplish. Also, the SRS contains definitions of technical terms so that the readers could comprehend all. The SRS also details the connection of the LMS with other systems in the vehicle and details each of its key components. Further, the SRS includes the various ways in which the users will be interacting with the system and rules that the system should abide by. On the same note, the SRS enumerates the safety features, potential future upgrades, and special conditions that need to be taken into consideration. Finally, the SRS has examples, diagrams, and how to test the features of the system.

1.1. Purpose

The SRS document is aimed at being a clear explanation of what the LMS is expected to do. The SRS also aids in making all involved people know what the system is meant to achieve, what it contains and how it is expected to act in various scenarios. Besides, the SRS ensures that the understanding is shared by all the team members, including developers, testers and project managers. Equally, customers and instructors will find it convenient to read and provide useful feedback in the design and building processes. This document serves as a common plan guide towards the creation and verification of the system at each stage. The SRS also eliminates confusion since the description of the system is done in a single location. The target audience will consist of software developers, testers, designers, project managers, customers, and instructors in the university.

1.2. Scope

The developed software product is known as the Lane Management System (LMS) that assists vehicles to remain safely in the road lanes. The LMS also contains three crucial components, Lane Departure Warning System, Lane Centering System, and Lane Keeping System, which are used to drive safely. Besides, the system provides cameras and sensors to identify the position of the vehicle on the road, and lane markings in real-time. This software enhances road safety, alerts drivers in case of lane drifting and even assists them in steering where necessary. Furthermore, the LMS is capable of more effective driving under various conditions of the road and weather with the help of smart technology [1]. The application area is a vehicle automotive safety embedded system which collaborates with the sensor and vehicle control systems. The system is installed in the vehicle and is capable of semi-automated driving.

The LMS will operate on the principle of integrating camera sensing, image processing, and movement data of vehicles to aid the control of lanes. The LMS will also come with systems on warning the driver, centering the vehicle mildly and active steering of the vehicle in its lane. Moreover, the LMS will not operate in all cases, such as the vehicle moving at a particular speed and when the information about lanes is not clearly visible. Besides, the system will also provide the driver with control when necessary, particularly in adverse or unclear roads. Likewise, the LMS will also deactivate in a safe way in case any key component has failed and will warn the driver. These characteristics correspond to the initial project requirements of the customer that are concerned with safety, automation, and appropriate application of sensor-based lane detection.

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1.3. Definitions, acronyms, and abbreviations

LMS: Lane Management System – the complete driver-assistance software system.

LDWS: Lane Departure Warning System – alerts the driver when the vehicle drifts.

LCS: Lane Centering System – applies minor corrections to keep the vehicle centered.

LKS: Lane Keeping System – actively steers the vehicle back into the lane if needed.

GPS: Global Positioning System – used to provide vehicle location coordinates.

CAN Bus: Controller Area Network – vehicle communication network for subsystems.

UI: User Interface – the driver's way of interacting with the LMS.

Subsystem: A smaller system that is part of the overall LMS architecture.

1.4. Organization

The remaining parts of this SRS document describe the Lane Management System in further detail in well-structured parts. It also contains a complete description of the system functionality, its key features, user roles, and limitations of the system. Also, the hardware and software specifications, interface, security guidelines, and behavior of the system in various circumstances are discussed. In addition, there are diagrams and examples used to support how the system would behave in real life. On the same note, the document enumerates all suppositions, safety regulations, and potential future modifications that could be undertaken. The SRS is structured into the following sections:

Section 1:Introduces the customer to the product and how this SRS is organized
Section 2: High level overview of the product
Section 3 : Lists the detailed hardware, software, and security requirements for the system
Section 4: Modeling requirements which includes diagrams for better understanding
Section5: Describes the prototype and overview at how the system will work.
Section 6: References in IEEE format
Section 7: Contact information for Dr. Betty Cheng

2. Overall Description

This section gives a comprehensive description of the LMS and gives the key design concepts, constraints, and user expectations. It also outlines the components involved, their functions, and the connection of the system to the other vehicle systems. Furthermore, this section demonstrates what the system is capable of, who will use it, and the requirements needed to make it work. Moreover, it contains information on software and hardware configuration, user expertise as well as environment requirements. It is supposed to guide the readers to gain knowledge of the system first before hearing the technical requirements.

2.1. Product Perspective

Lane Management System belongs to a broader vehicle safety platform and is integrated with other in-vehicle systems to enhance driving. It also links to devices such as vehicle speed detectors, cameras, steering control and driver alert systems. The LMS is also an individual unit within the vehicle which relies on sensor information and embedded software modules [2]. System interfaces are camera input, sensor data of the vehicle systems and steering feedback control. The user interfaces consist of a dashboard interface and system settings controls. Hardware connectors are interfaces, which unite the cameras, steering control and vehicle sensors. Vehicle operating system links are contained in software interfaces. All LMS parts have communication interfaces that enable the flow of data. The constraints are the memory usage, the safe operating instructions and the capability to make adjustments to the system where necessary.

2.2. Product Functions

The LMS will monitor the lane markings with the camera, and provide warnings in case the vehicle moves out of its lane without a signal. It will also give light steering assistance to ensure that the vehicle is in the center and more powerful steering assistance in case of failure by the driver to respond. Furthermore, it records valuable information such as the speed of the vehicle, steering angle, and GPS to monitor the events and assist in the future analysis. The LMS also has various levels of alerts to correspond to the level of danger and settings that are selected by the driver. These functions assist in minimizing the accidents, enhancing driver awareness, and semi-automated control of vehicles. The system collaborates with all the subsystems to identify, alert and modify vehicle movement.

2.3. User Characteristics

The primary customers the LMS will be ordinary drivers who travel in vehicles whether at a long distance or daily. These drivers should possess a valid driver's license, and simple instructions will be given to them regarding the use of LMS features. Also, the system can be used by vehicle technicians to set up and update, and thus, the technicians need to be familiar with system settings and safety regulations. In addition, drivers do not need technical knowledge, but must be able to follow warnings, alerts, and control settings with a lot of ease. The system is also supposed to be friendly to all drivers since it will have simple signals and controls. This makes the system usable by drivers of all ages, skills and driving experiences in a variety of conditions and environments.

2.4. Constraints

The LMS has some of the main limitations that have to be strictly adhered to maintain system integrity, ensure the safety of drivers, and provide certain reliability in its work; the set of limitations characterizes the scope of work of the system in different environmental, technical, and regulatory requirements and conditions and is vital in preventing unsafe behavior, maintaining driver control, and preventing risks related to sensor failure, low visibility, or poor weather conditions.

Unsafe steering actions should not be practiced by the LMS and should also be able to give back control to the driver when necessary. It should also be able to close safely in case a major system fails, or the lane markings are not very clear. Moreover, the LMS should not operate below the speed of 35 mph, and in the conditions of heavy rain, snow, fog, etc. to prevent mistakes. In addition, when there is a lack of memory or sensor data, the system is also not supposed to take action but provide an immediate warning to the driver. The program should be strict on safety and privacy regulations to ensure that no errors or attacks occur to the driver and the vehicle. They are very crucial constraints since failure in any of them can result in unsafe driving, wrong warnings, or steering issues and this should be avoided at all times.

2.5. Assumptions and Dependencies

The LMS relies on the installed cameras, sensors and steering controls in the vehicle. It should also have a dependable GPS, weather sensors and vehicle speed information to make the correct decisions in time. Also, the system presupposes that there are road lane markings which are correct and visible during the majority of the trip. Furthermore, it presupposes that drivers are not distracted and are not ignoring system feedback and will act upon warnings. The LMS software must operate on an automotive compatible system that can transmit updates, as well as store event data. LMS may not properly operate in case some sensor, camera, or control system is broken. Such dependencies are to be thoroughly tested and maintained to ensure the safety and the expected functionality of the system.

2.6. Apportioning of Requirements

According to the discussions with the customer, several features are not in the scope of the current version of the LMS but are also intended to be included in the future iterations. An example is multiple levels of alerts depending on the perceived risk of the event occurring such as compatibility with haptic, audio, and visual alerts. Additionally, detailed logging (such as storing information when errors occur or just in general consistently) can also be introduced in the future. Lastly, more detailed usage of the onboard artificial intelligence used to decode visual input is something that could be implemented later.

3. Specific Requirements

This section overviews the functional and non-functional specifications for the Lane Management System (LMS), ensuring clear subsystem boundaries, safety-critical behavior, and verifiable performance specifications.

3.1. Functional Requirements

3.1.1. Lane Detection and Tracking

3.1.1.1. The system shall continuously detect lane markings using the front-mounted camera positioned behind the rear-view mirror.

3.1.1.2. The system shall determine the host vehicle's position relative to the detected lane line (left, center, or right).

3.1.1.3. If no valid lane markings are detected, the system shall transition to *Standby Mode* until valid markings are reacquired.

3.1.1.4. The system shall maintain at least **95% confidence** in lane detection before issuing corrections or warnings.

3.1.1.5. The system shall use radar and machine-learning-based fusion to enhance lane detection under variable lighting and curvature conditions.

3.1.2. Lane Departure Warning System (LDWS)

3.1.2.1. LDWS shall issue a visual warning when the vehicle departs from the lane without a turn signal when the system is active.

- Each lane-departure warning event shall generate a single alert.
- repeated drifts shall trigger new warnings only after the first correction. The continued warnings increase in sound.
- Warnings shall include visual and haptic feedback, escalating to auditory alerts for severe cases of lane departure.

3.1.2.2. If lane departure is intentional (e.g., driver turn signal active), the lane departure warning shall be suppressed.

3.1.3. Lane Centering System (LCS)

3.1.3.1. LCS shall remain active at all times when valid lane have been identified by the detectors in the system and analyzed by the LCS.

3.1.3.2. The system shall issue a non-distracting light warning when approaching a lane boundary.

3.1.4. Lane Keeping System (LKS)

3.1.4.1. LKS shall activate when the host vehicle crosses or approaches a lane boundary beyond LCS control threshold, being contact with the lane boundry.

3.1.4.2. The system shall apply corrective steering torque to return the vehicle to the lane.

3.1.4.3. LKS shall immediately deactivate if the driver exerts sufficient counter-torque proportional to the misalignment, applies brakes, or engages turn signals.

3.1.4.4. LKS shall disable entirely below 35 mph.

3.1.5. Driver Interface and Control

3.1.5.1. The driver shall be able to enable or disable LMS through a global on/off switch.

3.1.5.2. The system shall use two alert levels (Low Level Alert, and High Level Alert) corresponding to increasing lane-departure severity.

- The system shall issue a Low Level Warning alert when the vehicle begins to drift toward a lane boundary but remains within the lane, indicating early off-center deviation.

- The system shall issue a High Level Alert when the vehicle has left the immediate bounds of the lane it is positioned within.

3.1.5.3. Upon vehicle ignition, the LMS shall resume its last known operational state.

3.1.6. System Integration and Supervisory Control

3.1.6.1. The LMS shall integrate the camera sensing subsystem for raw image capture.

- The camera subsystem shall output timestamped image frames to the image-processing subsystem.

3.1.6.2. The LMS shall integrate the image-processing subsystem for lane-feature extraction.

- The image-processing subsystem shall output detected lane markings and confidence values to the vehicle-state-estimation subsystem.

3.1.6.3. If any subsystem or hardware module fails, the LMS shall shut off and notify the driver via visual and auditory alerts.

3.1.6.4. Upon subsystem recovery, LMS shall automatically restore the previous active state, while notifying the driver.

3.1.7. Environmental Adaptation

3.1.7.1. The LMS shall detect adverse weather using vehicle temperature and water sensors, this being temperatures below 0 degrees Celsius and consistent precipitation lasting for longer than 10 minutes

3.1.7.2. Under heavy rain, snow, or fog, the LMS shall transition to standby mode to prevent unsafe operation.

3.1.7.3. In construction zones or any area with conflicting markings due to temporary markings or self intersection lanes, the LMS shall disable and alert the driver.

3.1.8. Logging and Data Management

3.1.8.1. The system shall log each warning correction, and subsystem fault event.

3.1.8.2. Each log entry shall include vehicle speed, steering angle, and GPS location.

3.1.8.3. Logged data shall comply with all applicable regulatory privacy and data-usage standards.

3.2. Non-Functional Requirements

3.2.1. Safety and Reliability

3.2.1.1. LMS shall fail-safe, reverting to driver control in all fault conditions.

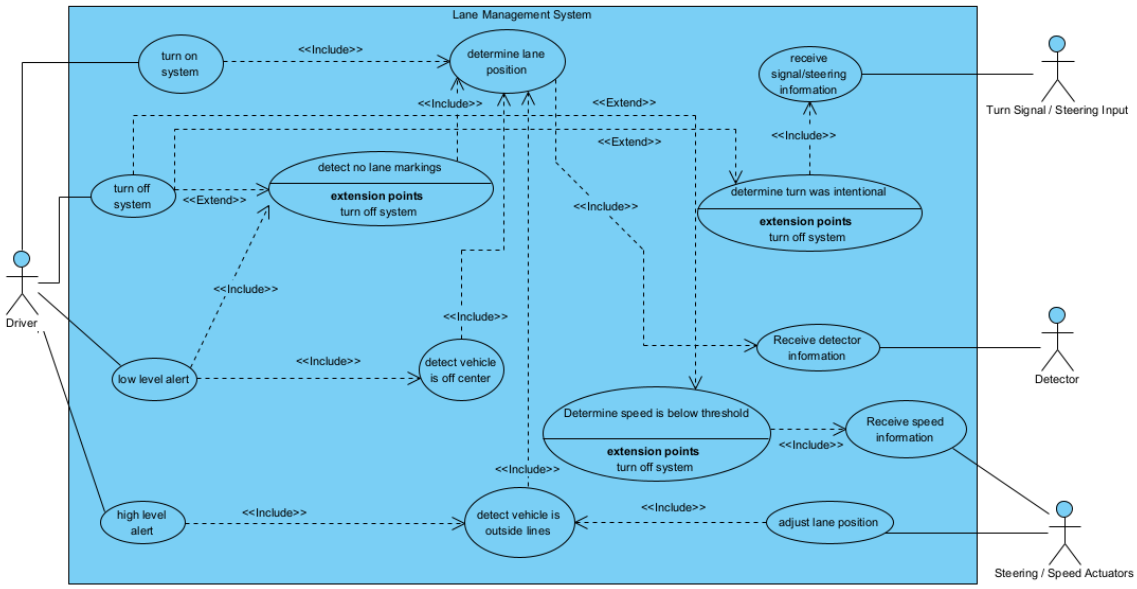
4. Modeling Requirements

This section demonstrates the use cases of our system, including a use case diagram, domain model, three sequence diagrams, and four state models.

4.1. Use Case Diagrams and Descriptions

The following subsections describe and illustrate a use case diagram that covers all requirements. A detailed description of the model, and a series of tables are included in section 4.1.2 for an explanation of each use case present in the diagram.

4.1.1. Use case Diagram



4.1.2. Use Case Descriptions

Use Case:	*Turn on system*
Actors:	*Driver*
Description:	*The driver presses a physical button present on the vehicle's central console. From then on, until the driver turns off the system, all 3 major subsystems are activated and attempting to detect lane markings.*
Type:	*Primary (essential)*
Includes:	*Determine lane position*
Extends:	N/A
Cross-refs:	*3.1.5.3, 3.1.6.4*
Use cases:	*Determine lane position*

Use Case:	*Turn off system*
Actors:	*Driver*
Description:	*The driver presses a physical button present on the vehicle's central console or no available lane markings are present. From then, all 3 major subsystems are de-activated and a low level warning is displayed.*
Type:	*Primary (essential)*
Includes:	*Detect vehicle is off center*
Extends:	N/A
Cross-refs:	3.1.5.1, 3.2.1.1
Use cases:	*Detect no lane markings, Detect vehicle is off center, Determine Turn was Intentional, Determine Speed is Below Threshold*

Use Case:	*Determine Lane position*
Actors:	*N/A*
Description:	*The LMS determines the vehicle's location on the road relative to the road markings received from the Detector actor.*
Type:	*Primary (essential)*
Includes:	*Receive Detector Information*
Extends:	N/A
Cross-refs:	*3.1.1.1, 3.1.1.2, 3.1.8.1, 3.1.3.1, 3.1.8.2, 3.1.1.5, 3.1.1.4, 3.1.8.3, 3.1.1.3, 3.1.6.2*
Use cases:	*Receive Detector Information, Detect Vehicle is Outside Lines, Detect Vehicle is Off Center, Detect No Lane Markings, Turn On System*

Use Case:	*Receive Detector Information*
Actors:	*Detector*
Description:	*The front-mounted camera takes a photo of what is in front of the vehicle at the current time and processes the information.*
Type:	*Primary (essential)*
Includes:	*N/A*
Extends:	N/A
Cross-refs:	*3.1.6.1, 3.1.6.2*
Use cases:	*Determine Lane Position*

Use Case:	*Detect No Lane Markings*
Actors:	*N/A*
Description:	*The LMS determines that the vehicle's relative location on the road cannot be compared against any known lane markings.*
Type:	*Primary (essential)*
Includes:	*Determine Lane Position*
Extends:	Turn Off System
Cross-refs:	3.1.6.3, 3.1.7.1, 3.1.7.2, 3.1.7.3
Use cases:	*Turn Off System, Low Level Alert, Determine Lane Position*

Use Case:	*Detect Vehicle is Off Center*
Actors:	*N/A*
Description:	*The LMS determines the vehicle's location on the road is outside of the center of the lane.*
Type:	*Primary (essential)*
Includes:	*Determine Lane Position*
Extends:	N/A
Cross-refs:	3.1.3.2
Use cases:	*Low Level Alert, Determine Lane Position*

Use Case:	*Low Level Alert*
Actors:	*Driver*
Description:	*The system utilizes a passive alert to notify users that either the system must be temporarily de-activated, or they are veering outside of the center of the lane.*
Type:	*Primary (essential)*
Includes:	*Detect No Lane Markings, Detect Vehicle is Off Center*
Extends:	N/A
Cross-refs:	3.1.3.2, 3.1.5.2
Use cases:	*Detect No Lane Markings, Detect Vehicle is Off Center*

Use Case:	*Detect Vehicle is Outside Lines*
Actors:	*N/A*
Description:	*The LMS determines the vehicle's location on the road is outside of the visible road lines.*
Type:	*Primary (essential)*
Includes:	*Determine Lane Position*
Extends:	N/A
Cross-refs:	3.1.2.1, 3.1.4.1
Use cases:	*Adjust Lane Position, High Level Alert, Determine Lane Position*

Use Case:	*Adjust Lane Position*
Actors:	*Steering Actuators*
Description:	*The steering wheel is given orders to adjust the angle that the vehicle is moving in. This action is taken to directly counteract the vehicle's drifting out of the lane it was previously full inside of.*
Type:	*Primary (essential)*
Includes:	*Detect Vehicle is Outside Lines*
Extends:	N/A
Cross-refs:	3.1.4.2,
Use cases:	*Detect Vehicle is Outside Lines*

Use Case:	*High Level Alert*
Actors:	*Driver*
Description:	*A louder and more apparent alert is displayed to the driver to let them know that the LKS subsystem is taking action to steer their vehicle back into the lane.*
Type:	*Primary (essential)*
Includes:	*Detect Vehicle is Outside Lines*
Extends:	N/A
Cross-refs:	3.1.2.2, 3.1.5.2
Use cases:	*Detect Vehicle is Outside Lines*

Use Case:	*Receive signal/steering information*
Actors:	*Turn Signal / Steering Input*
Description:	*The system receives data from the turn signal and steering wheel regarding their status/angle.*
Type:	*Primary (essential)*
Includes:	*N/A*
Extends:	N/A
Cross-refs:	*3.1.2.2*
Use cases:	*Determine Turn Was Intentional*

Use Case:	*Determine Turn Was Intentional*
Actors:	*N/A*
Description:	*The system digests turn signal/steering data to determine if the user is exerting enough torque to qualify as an intentional turn or if their turn signal is active.*
Type:	*Primary (essential)*
Includes:	*Receive signal/steering information*
Extends:	Turn Off System
Cross-refs:	3.1.2.2, 3.1.4.3
Use cases:	*Receive signal/steering information, Turn Off System*

Use Case:	*Receive Speed Information*
Actors:	*Steering/Speed Actuators*
Description:	*The system pulls data from the steering wheel to determine how fast the vehicle is traveling.*
Type:	*Primary (essential)*
Includes:	*N/A*
Extends:	*N/A*
Cross-refs:	3.1.4.4
Use cases:	*Determine Speed is Below Threshold*

Use Case:	*Determine Speed is Below Threshold*
Actors:	*N/A*
Description:	*The system utilizes the data from the speedometer to determine it should remain online given the vehicle's current speed.*
Type:	*Primary (essential)*
Includes:	*Receive Speed Information*
Extends:	Turn Off System
Cross-refs:	3.1.4.4
Use cases:	*Receive Speed Information, Turn Off System*

4.2. Domain Model and Data Dictionary

The following domain model illustrates the system's intended class structure. Each of the main blue boxes present in the diagram describes the attributes and operations of a given class alongside that attribute or operation's visibility to other classes. White diamond connectors indicate an aggregation, black diamond connectors indicate a composition, and white arrow connectors indicate an inheritance.

4.2.1. Domain Model

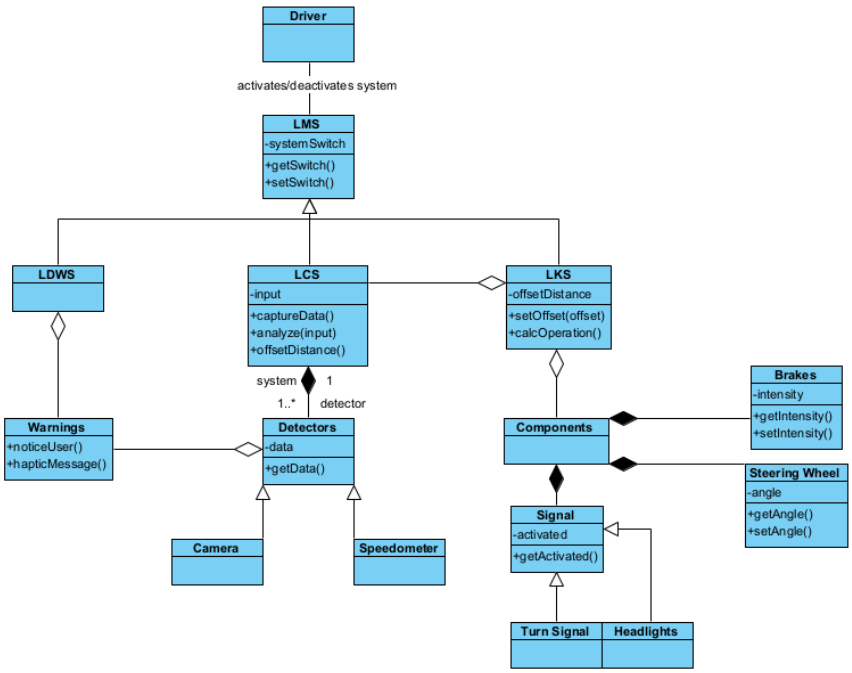


Figure 1: High level domain model for LMS

4.2.2. Data Dictionary

Included below is the data dictionary of the domain model above. It describes each class in detail, its relationship with other classes (included notes about any present aggregations, compositions, or inheritances), its attributes, and its operations.

Element Name		**Description**
Driver		The class that represents the user driving the vehicle.
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	LMS : Association	

Element Name		Description
LMS		The class that represents the entire LMS System
Attributes		
	systemSwitch	A Boolean that determines whether LMS is active or offline.
Operations		
	getSwitch()	Retrieves the status of the systemSwitch variable
	setSwitch(switch Boolean)	Sets the systemSwitch variable to true or false based on the passed variable
Relationships	LMS has 3 subsystems that all maintain compositions with it. LDWS has a composition with LMS, LCS has a composition with LMS, and LKS has a composition with LMS.	

Element Name		Description
LDWS		The class that represents the Lane Detection Warning System aspect of LMS
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	LDWS is a composition of LMS and aggregates the Warnings class.	

Element Name		Description
LCS		The class that represents the Lane Centering System.
Attributes		
	input	Stores both the speedometer data received from the speedometer and the processed camera data.
Operations		
	getData()	Retrieves the data from each connected Detector.
Relationships	Detectors is a composition of LCS and is composed of LMS. LCS is also an aggregation of LKS.	

Element Name		Description
LKS		The class that represents the Lane Keeping System
Attributes		
	offsetDistance	Stores the distance the vehicle is away from the center lane.
Operations		
	setOffset(offset)	Sets the offset variable
	calcOperation()	Calculates the angle the steering wheel should take to correct the vehicle's angle.
Relationships	Components is an aggregation of LKS. LCS is an aggregation of LKS. LKS inherits from LMS.	

Element Name		Description
Warnings		The class that represents the entire warnings displayed to the user
Attributes		
	N/A	N/A
Operations		
	noticeUser()	Alerts the user that a warning is occurring
	hapticMessage()	Alerts the user via haptic feedback.
Relationships	Warnings is an aggregation of LDWS and Detectors.	

Element Name		Description
Detectors		The class that represents the detectors inputting sensory data into the system.
Attributes		
	data	Stores the data of the detector type this detector is representing
Operations		
	getData()	Retrieves the data being stored in the data variable.
Relationships	Camera and Speedometer inherit from Detectors. Warnings is an aggregation of Detectors and Detectors is a composition of LCS.	

Element Name		Description
Components		The class that represents the components that input data or are changeable by the system
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	Components is an aggregation of LKS and is composed of Brakes, Steering Wheel, and Signal	

Element Name		Description
Camera		The class that represents the Camera input system
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	Camera inherits from Detectors.	

Element Name		Description
Speedometer		The class that represents the Speedometer input system
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	Speedometer inherits from Detectors	

Element Name		Description
Signal		The class that represents the various types of signals present within the system
Attributes		
	activated	A Boolean that determines whether the given signal is active or not
Operations		
	getActivated()	Retrieves the status of the activated variable
Relationships	Signal is a composition of Components and both Turn Signal and Headlights inherit from Signal.	

Element Name		Description
Brakes		The class that represents the vehicle's brakes
Attributes		
	intensity	A double indicating how activated the brakes on the given vehicle are.
Operations		
	getIntensity()	Retrieves the status of the intensity variable
Relationships	Brakes are a composition of Components.	

Element Name		Description
Steering Wheel		The class that represents the vehicle's steering system
Attributes		
	angle	A double representing the angle that the steering wheel is currently at.
Operations		
	getAngle()	Retrieves the status of the angle variable
	setAngle(angle)	Sets the angle of the steering wheel to the angle provided as a parameter.
Relationships	Steering wheel is a composition of Components.	

Element Name		Description
Turn Signal		The class that represents the turn signal input into the system
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	Turn Signal is an inheritance of Signal	

Element Name		Description
Headlights		The class that represents the headlights input.
Attributes		
	N/A	N/A
Operations		
	N/A	N/A
Relationships	Headlights is an inheritance of Signal	

4.3. Sequence Diagrams

Several sequence diagrams displaying user interactions with the LMS are included in the following subsections. While they range from trivial to crucial in terms of system functionality, they all remain essential routines in the standard operation of the LMS.

4.3.1. Turning on the System

The following figure demonstrates the class interactions that occur when a user enables the LMS via a physical button present on the console. First, the system sets the "switch" boolean of the LDWS and LCS classes to "true". Next, the LCS class calls a function present within the Camera class and waits for a response containing the image data that the camera has most recently witnessed. Then, the LCS class calls upon the Speedometer, Brakes, Turn Signal, and Steering Wheel before analyzing if the system should remain active within the function "calcOperation()". If the system cannot remain

active, it pushes an alert to the LDWS class which displays an alert to the user via the Warnings class.

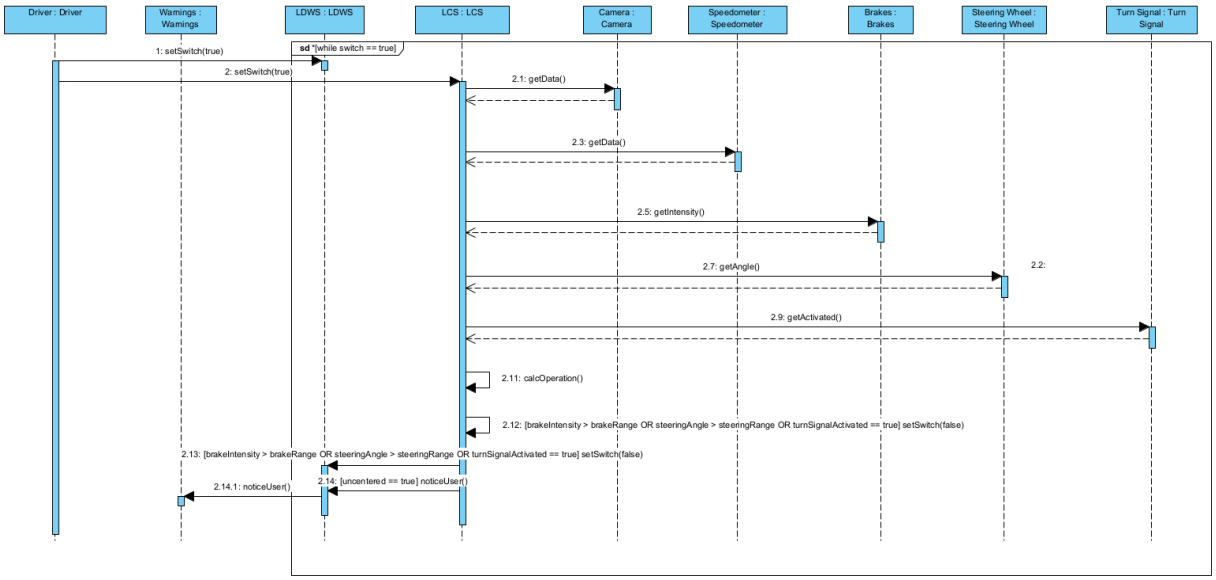


Figure 2: Sequence diagram relating to enabling the system

4.4. System activates LKS

The figure below demoinstrates what occurs when the system decides that LKS should take action towards steering the vehicle back within the marked road lines. LKS and LCS are both send a signal to activate, and LCS retrieves data from the Camera and the Speedometer. It then sends this data to LKS via the “setOffset()” function call. Next, LKS retrieves data from the brakes, steering wheel, and turn signal, and uses the function “calcOperation()” to determine whether the vehicle’s steering should be adjusted. If this is found to be the correct

path, the function “setAngle()” is utilized to reorient the direction of the vehicle, and the function “noticeUser()” is called to alert the driver that the vehicle is adjusting itself.

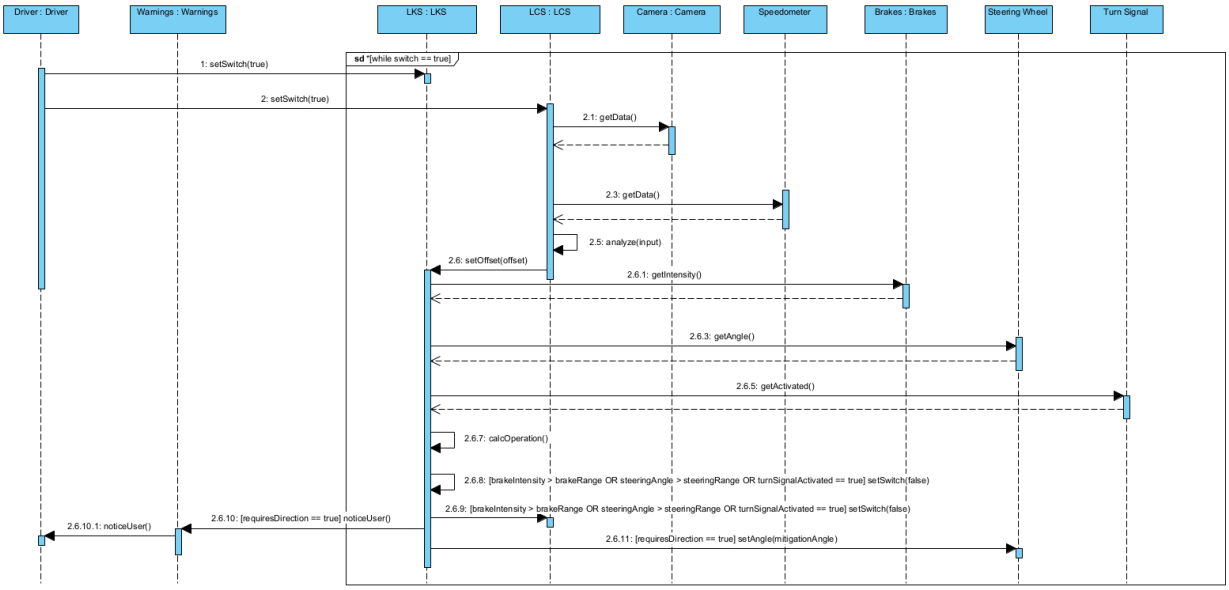


Figure 3: Sequence diagram relating to activating the Lane Keeping System

4.3.3. Camera is unavailable/weather is suboptimal for LMS

In the case that the camera is unavailable or the weather makes consistently identifying lane markings impossible, first the system powers on. Then, the LCS system determines that the camera’s data is unusable and sends a signal to disable the LDWS system and itself. Lastly, it sends a “noticeUser()” call through LDWS to the Warnings class so that the user is alerted to the system’s shutting off.

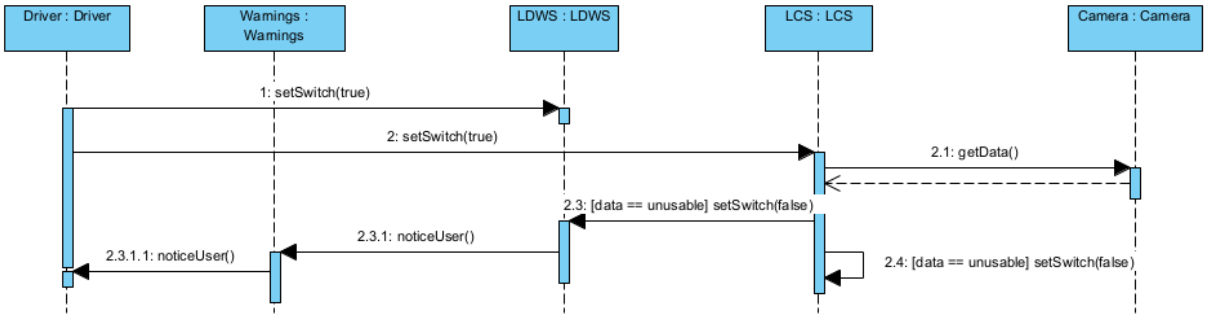


Figure 4: Sequence diagram relating to temporarily disabling LMS due to bad camera data or suboptimal weather conditions

4.4. State Diagrams

The following section describes the specific logic occurring to determine that state of relevant classes present within the aforementioned domain model. This information is in the form of state diagrams that allow the states of objects to be visibly displayed.

4.4.1. LCS Subsystem

The following figure displays the 3 states that the LCS can be in. It starts its lifetime in the idle state, then if it receives information that the vehicle is out of the center of the lane, it switches its state to "alerting user". Once the vehicle re-enters the center of the lane this state switches back to idle. The LCS will become de-activated if the system shuts down for any reason, and will always return to the idle state regardless of the state it was previously in before the shutdown.

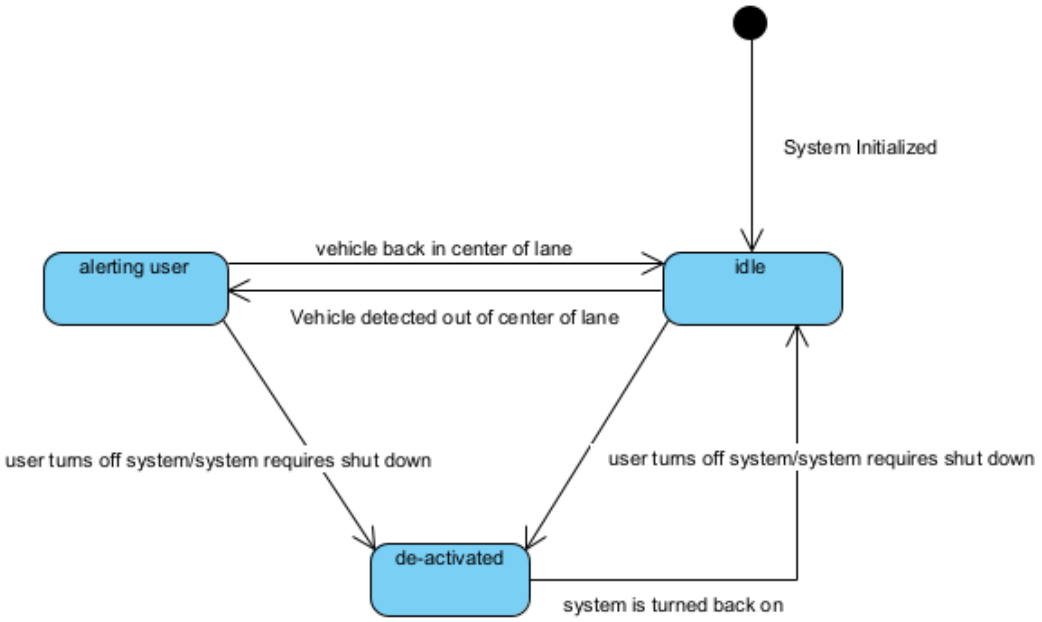


Figure 5: State diagram representing the LCS class

4.4.2. LDWS Subsystem

The following figure describes the possible states of the LDWS. It is chiefly similar to the LCS in terms of the number of potential states, however LDWS only switches to the "Alerting User" state upon fully exiting the perceived lane markings.

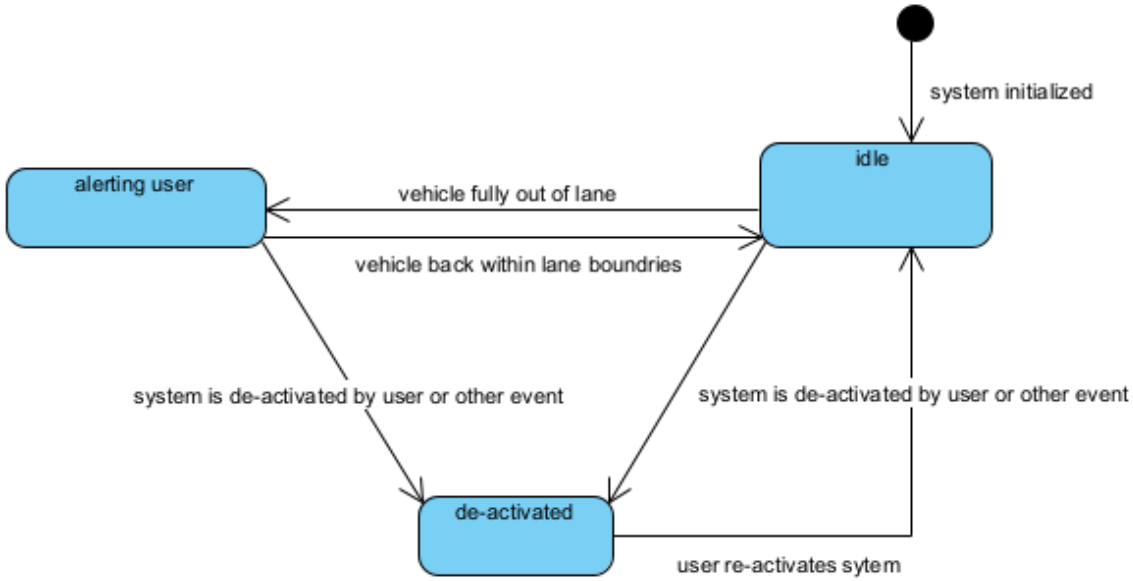


Figure 6: State diagram describing the LDWS class

4.4.3. LKS Subsystem

The following model details the potential states of the LKS. As with the LDWS and LCS, its execution is mainly defined by 3 main states: idle, de-activated, and alerting user/steering vehicle. The LKS is the only system that can steer the vehicle, so once it detects a lane departure, it switches its state to “steering vehicle”.

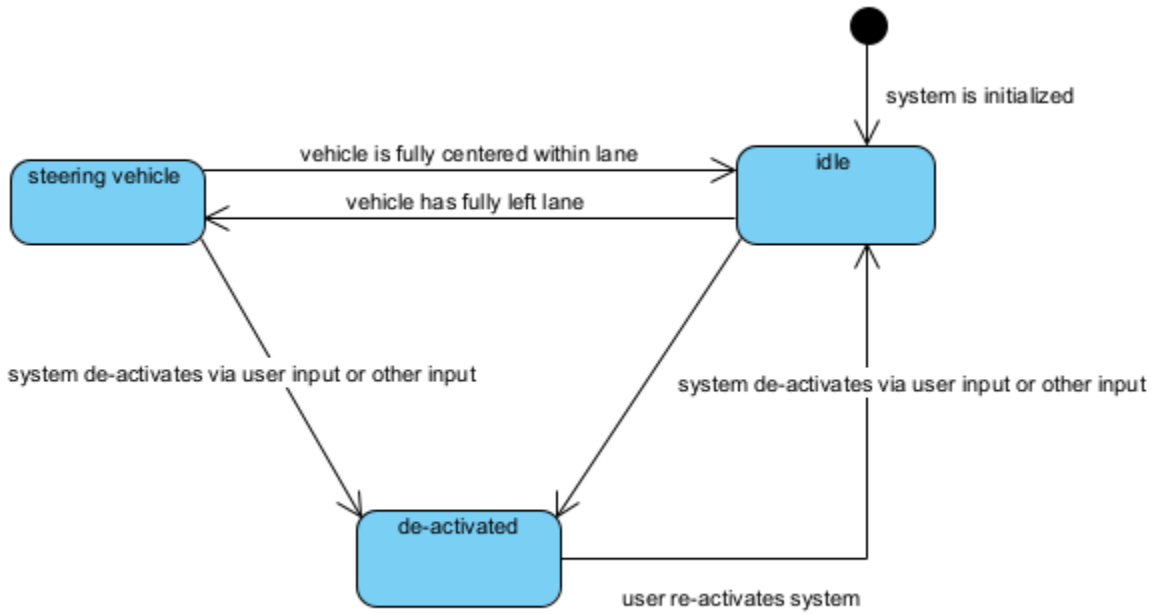


Figure 7: State diagram describing the LKS class

5. Prototype

This prototype will display the functionality of alerting the driver when they are outside the center of the lane, alerting the driver when they are outside the perceived lane markings, and intervening when the user has left their lane to steer them back into it. The following subsections will each explain how the prototype handles a specific system functionality. They will include captioned figures to assist in visualizing the scenario.

5.1. How to Run Prototype

A web browser is required to run the prototype. There are no plugins required or networking constraints. There are checkboxes present to select either an ideal scenario, a non-zero curvature scenario, or an absent lane information scenario. The experience is controllable via mouse motion. Moving the mouse towards the right will make the vehicle begin driving towards the right side of the screen. Simply click on the following link to experience the prototype:

https://play.unity.com/en/games/65252bb3-cd4e-4f5f-a277-d9f1335b10aa/cse-435-lms2-prototype

5.2. The Ideal Scenario

A driver is inattentive at the wheel traveling at 40mph on a straight road, and as a result, the driver begins to drift towards the left (See Figure 8 below). Once the vehicle leaves the center of the lane, the LCS system applies an alert to the driver (See Figure 9 below). Once the vehicle leaves the boundary of the lane, the LDWS system applies an alert to the driver (See Figure 10 below) and the LKS system steers the vehicle back into the proper lane (See Figure 11 below).

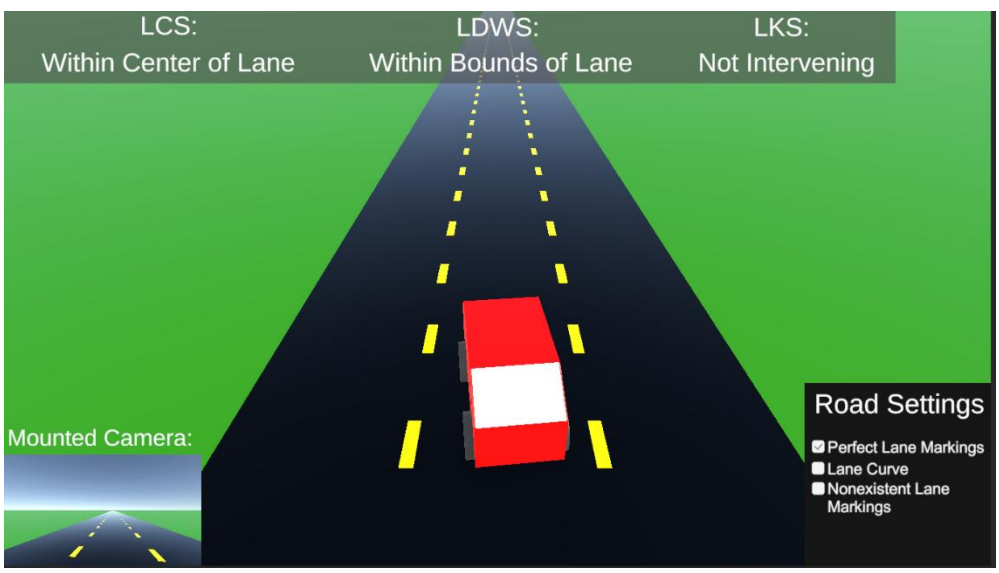


Figure 8: Driver begins to veer towards the left.

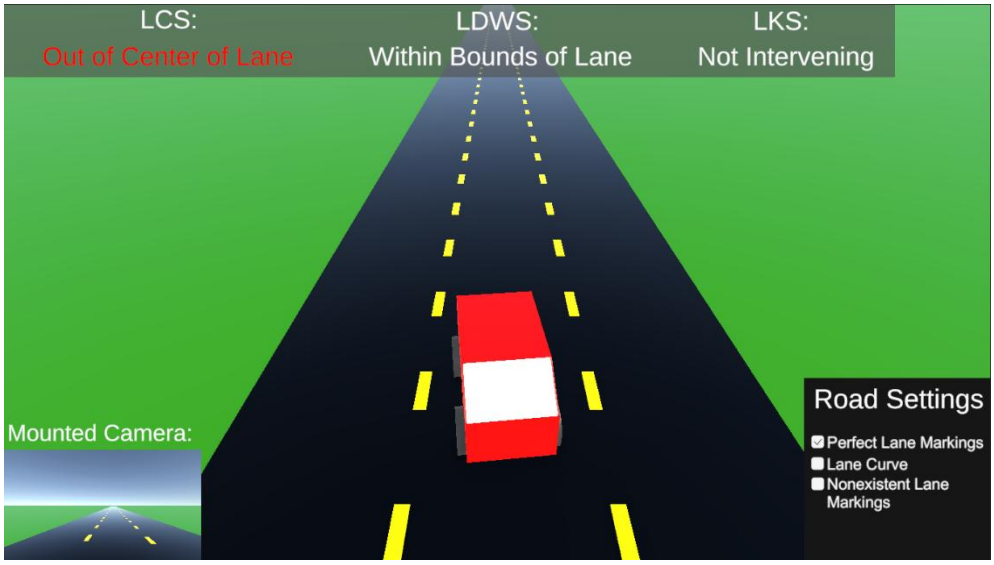


Figure 9: LCS alert activates to make the driver aware of their leaving the center of the lane.

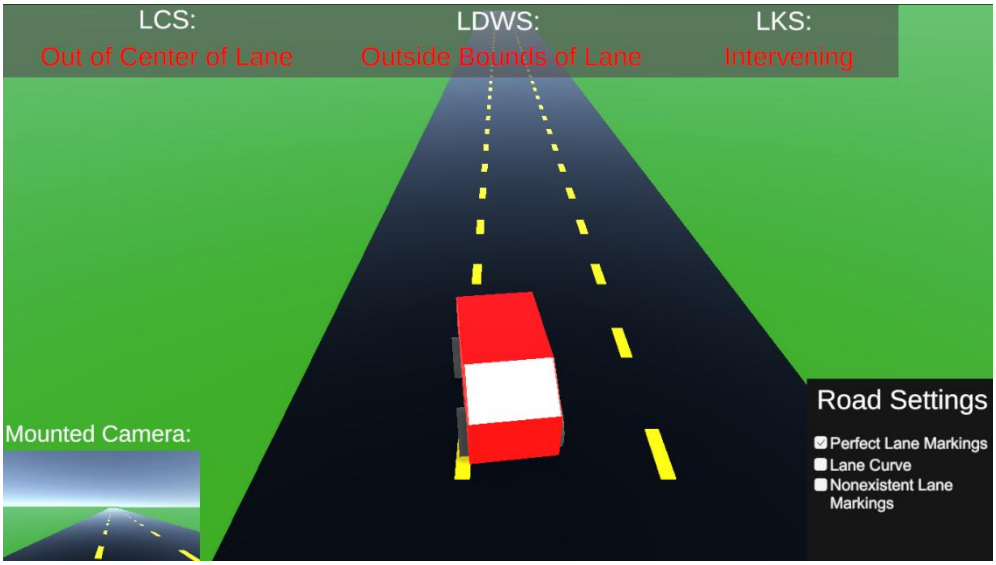


Figure 10: LDWS alert activates to make the driver aware that they have left their lane boundary and LKS activates to intervene and adjust the vehicle direction to steer back into the lane.

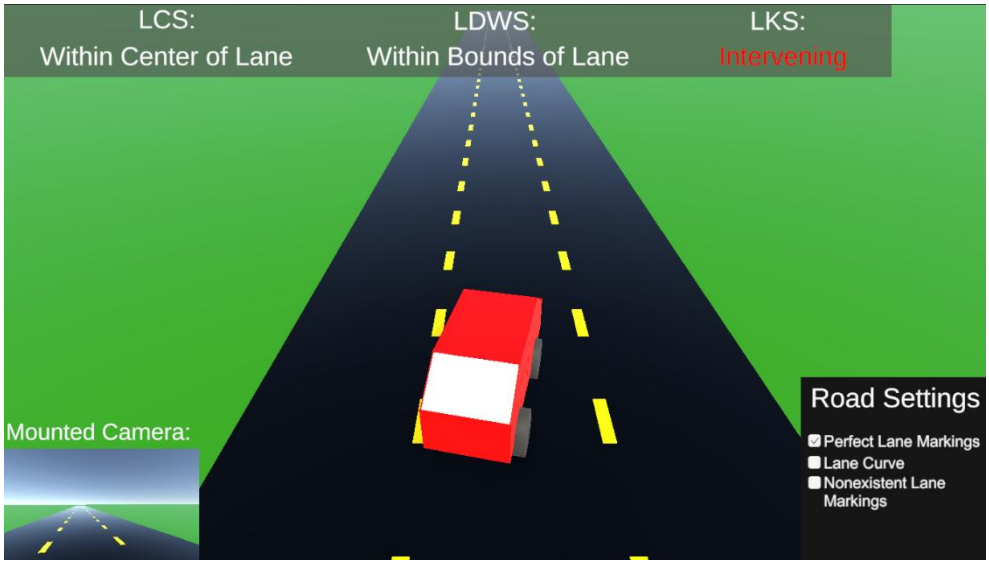


Figure 11: LKS steers the vehicle back into the lane.

5.3. Non-Zero Curvature

A driver becomes drowsy while driving and falls into a state of semi-consciousness. The driver remains in this state until a curved section in the road appears ahead (see Figure 12 below). Despite the severity of the curvature of this section of the road, the LCS provides a low level warning that the vehicle is drifting out of the center of the lane (see Figure 13 below). After the driver still does not regain full consciousness, the vehicle begins to drift out of the lane markings. At this time, LDWS issues a high level warning, and LKS begins intervening to reorient the vehicle back into the center of the lane (see Figure 14 below). After a moment of hearing the high level warning, the driver fully wakes up and jolts their vehicle the rest of the way back into the center of the lane with the assist of LKS (see Figure 15 below). After this occurs, the vehicle is now directly centered in the lane and as a result, LCS and LDWS cease warning the driver. In addition, LKS stops intervening to center the vehicle (see Figure 16 below).

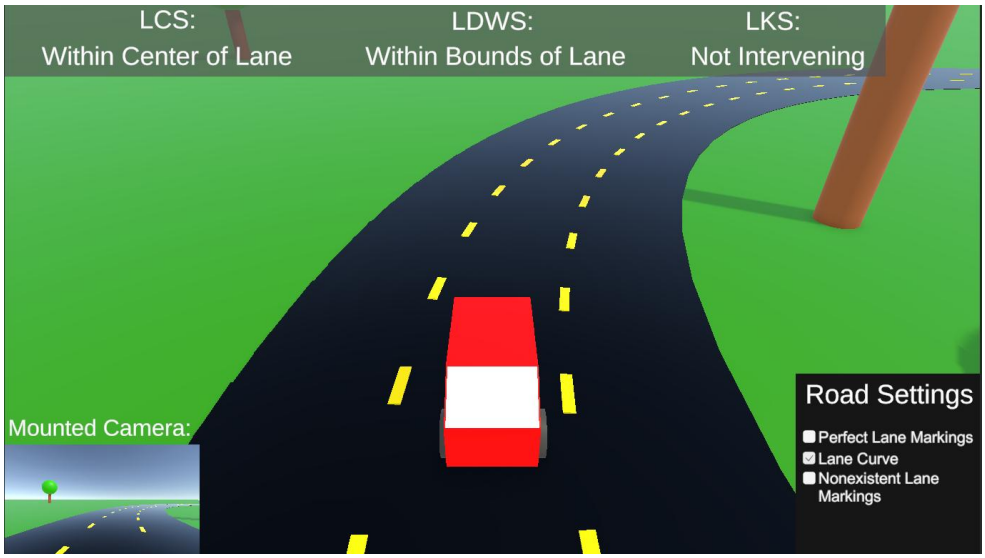


Figure 12: The vehicle remains within the center of the lane, however it has begun to drift out of the lane's center.

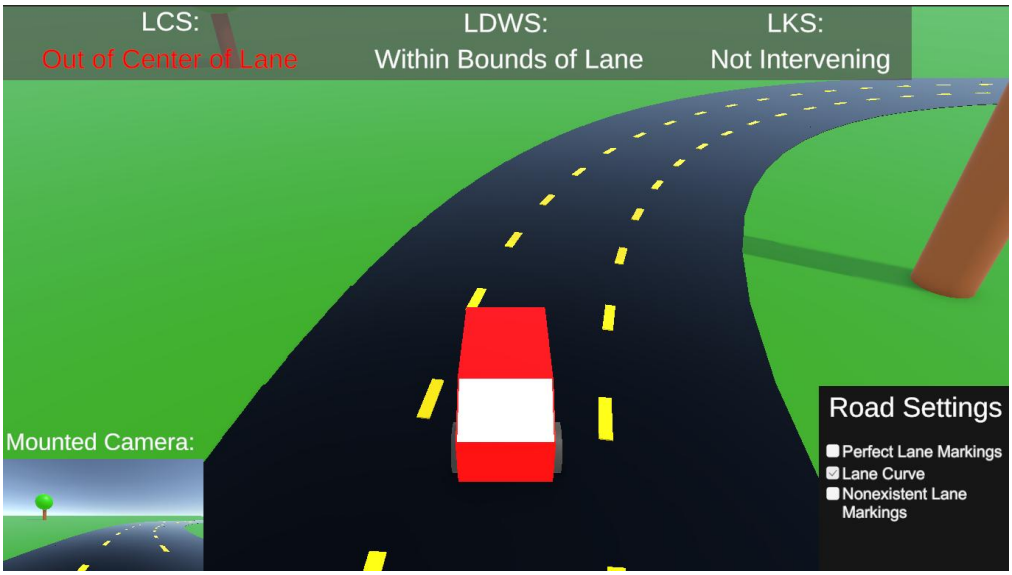


Figure 13: The vehicle has left the center of the lane, and LCS has began administering a low level warning to the driver.

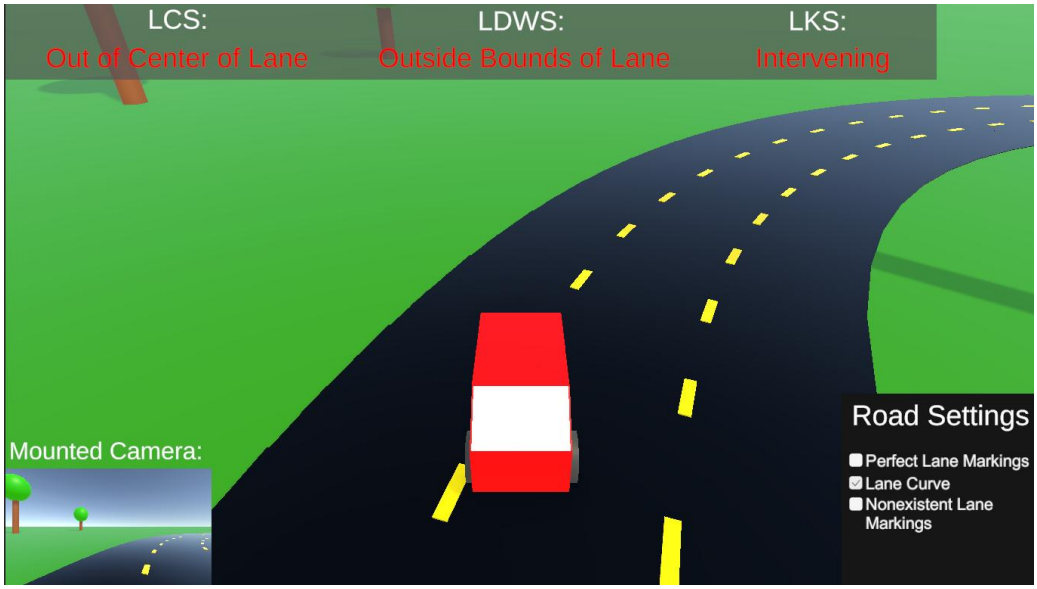


Figure 14: The vehicle has fully left the lane bounds, and in response, LDWS has begun administering a high level alert to the driver and LKS has begun intervention.

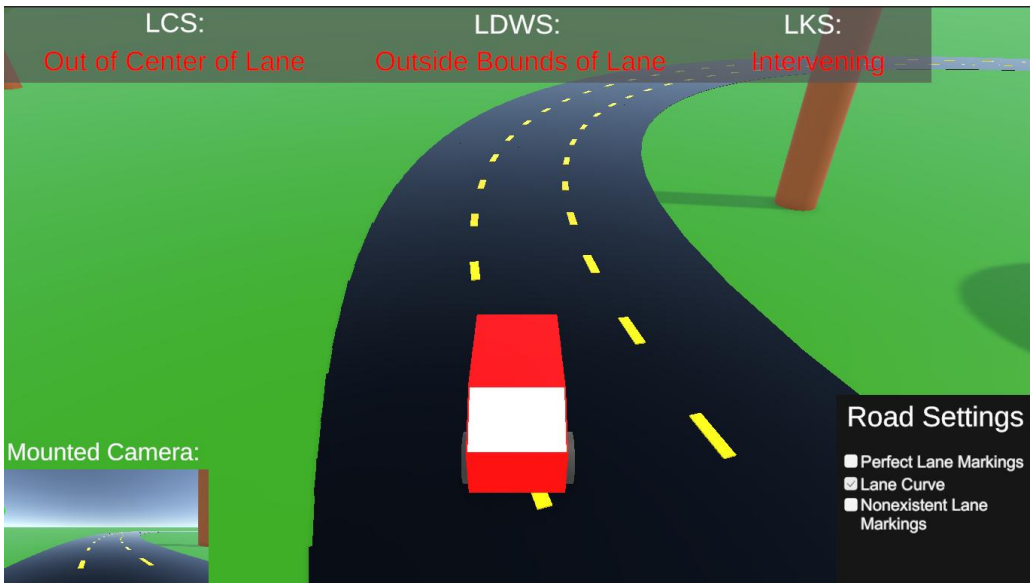


Figure 15: LKS aggressively steers the vehicle away from outside its lane to align it back to the center of its lane.

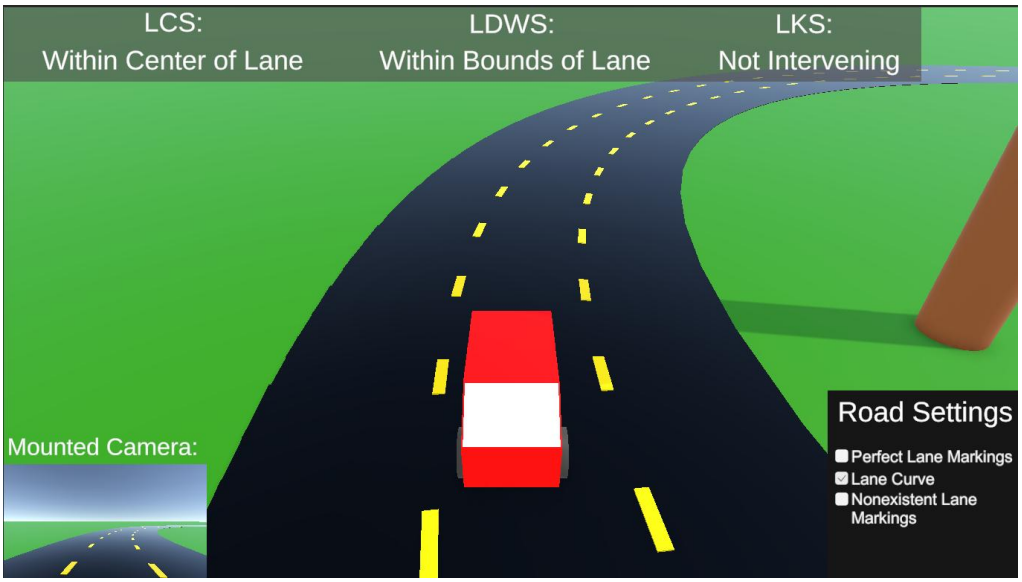


Figure 16: The vehicle has now safely been returned to the center of its lane and in response, LCS and LDWS have ceased sending warning messages to the driver. In addition LKS is no longer intervening.

5.4. Failure Scenario

While driving with LMS enabled on a road with clear lane markings, the user turns into a road without clear lane markings. As a result, the system is unable to attain valid lane state data and LMS becomes temporarily de-activated (See Figure 17 below). The user is given a low level warning, and the system attempts to restart once the lane markings become present again (see Figure 18 below).

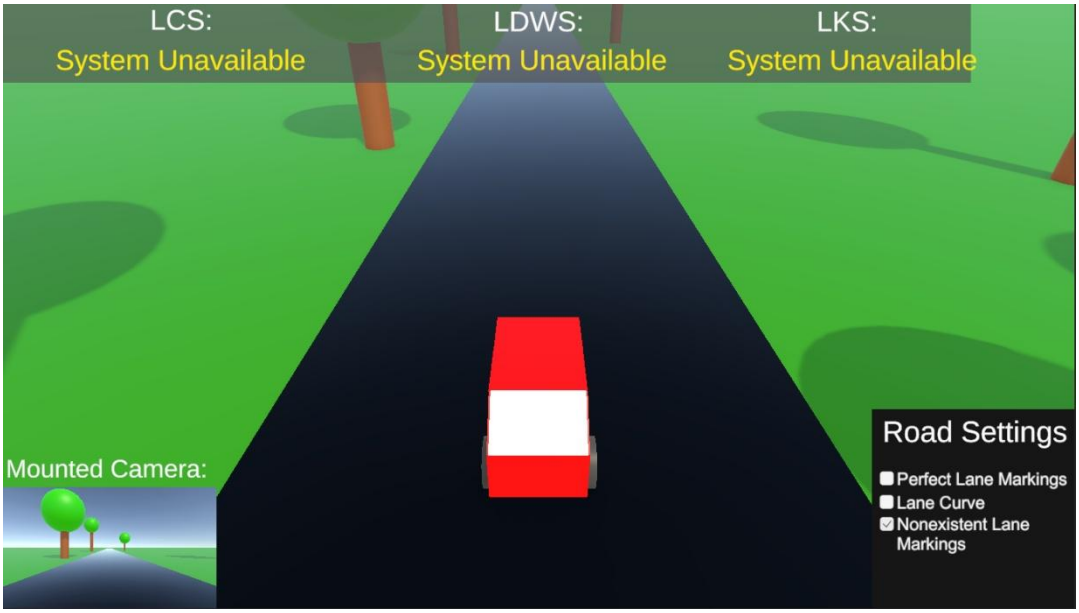


Figure 17: The system disables all main subsystems once data from the front mounted camera is absent/confusing.

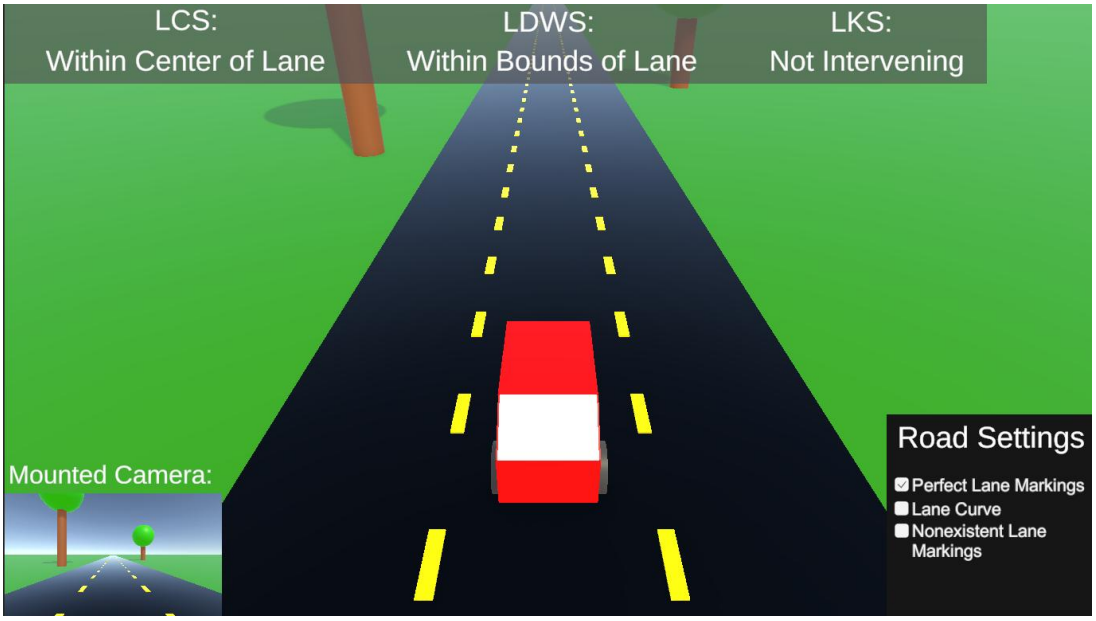


Figure 18: The system re-initializes without further user input once data from the camera is no longer confusing and line markings are clear again.

5.5. Complex Scenario

A vehicle is traveling at a high speed when the driver spots a large object lying in the lane ahead of them. The vehicle is traveling too fast to brake in time, so the driver jolts their wheel to the right to temporarily switch lanes (see Figure 19 below). Since the driver turned the wheel aggressively enough, LKS refused to intervene because it detected that the turn was intentional. The driver stays in the right lane until the large object passes (see Figure 20 below) and then once it does, the driver returns to the center lane (see Figure 21 & 22). Once the driver is fully returned the center lane, they have avoided a suddenly approaching terrain hazard without the LMS override their control.

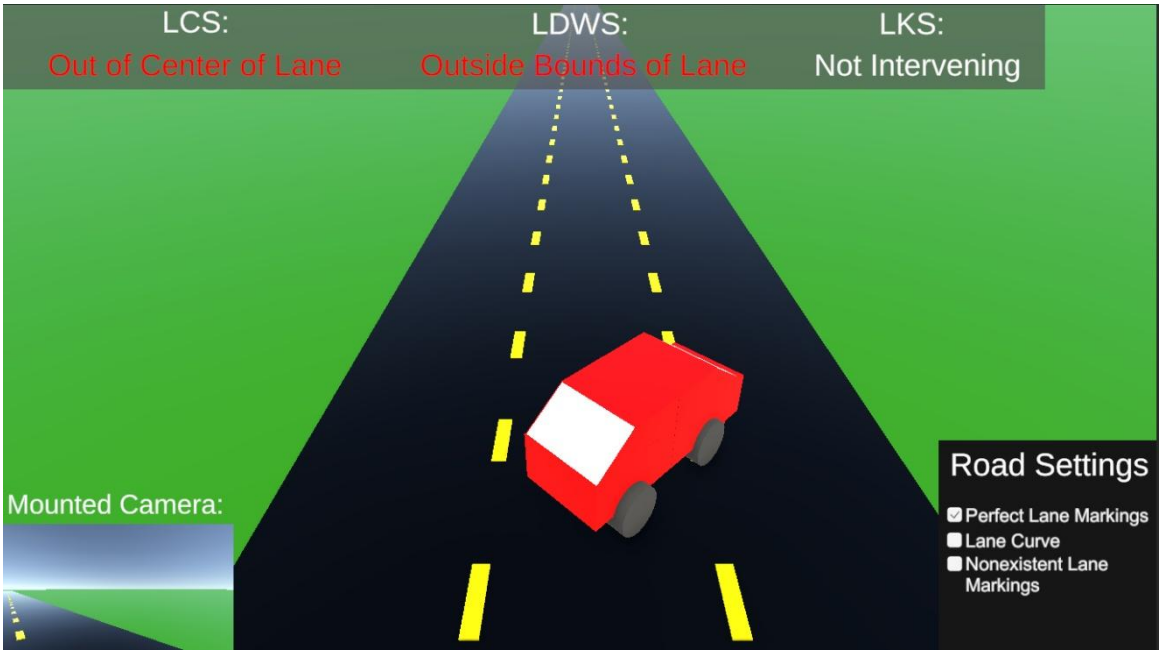


Figure 19: The driver jolts their steering wheel to the right to avoid the large object in the middle lane.

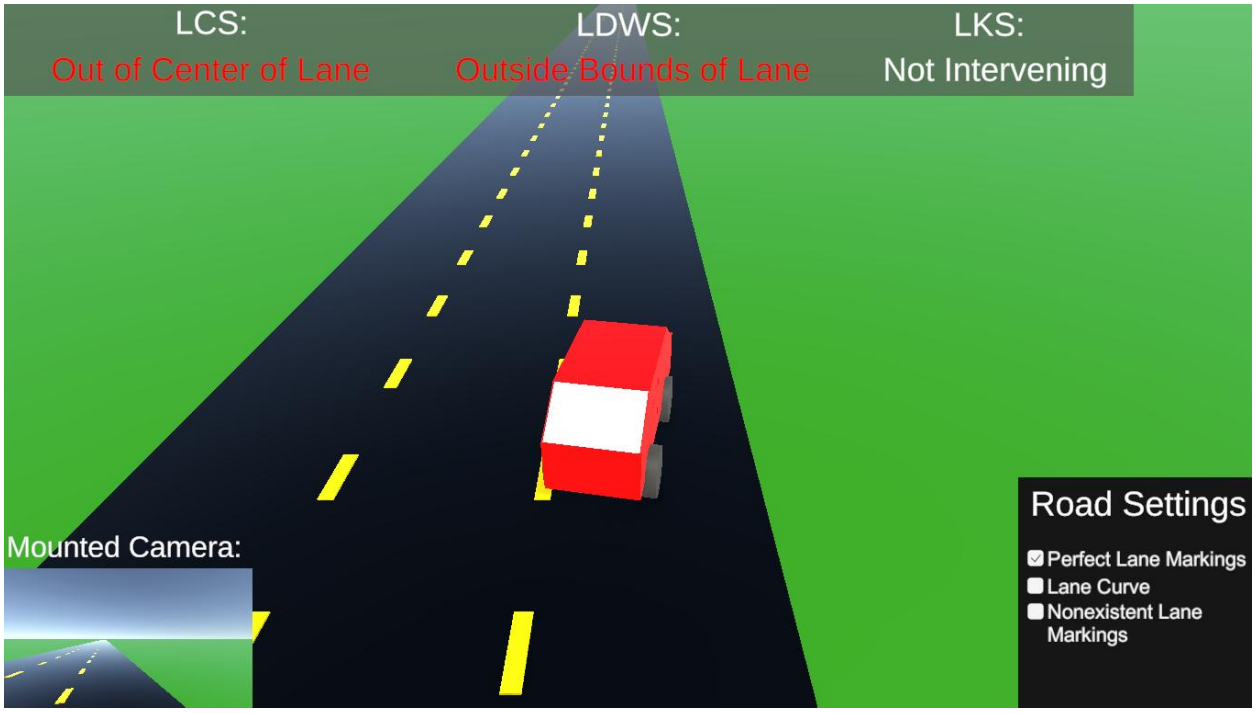


Figure 20: The driver remains in the right lane until the object has passed.

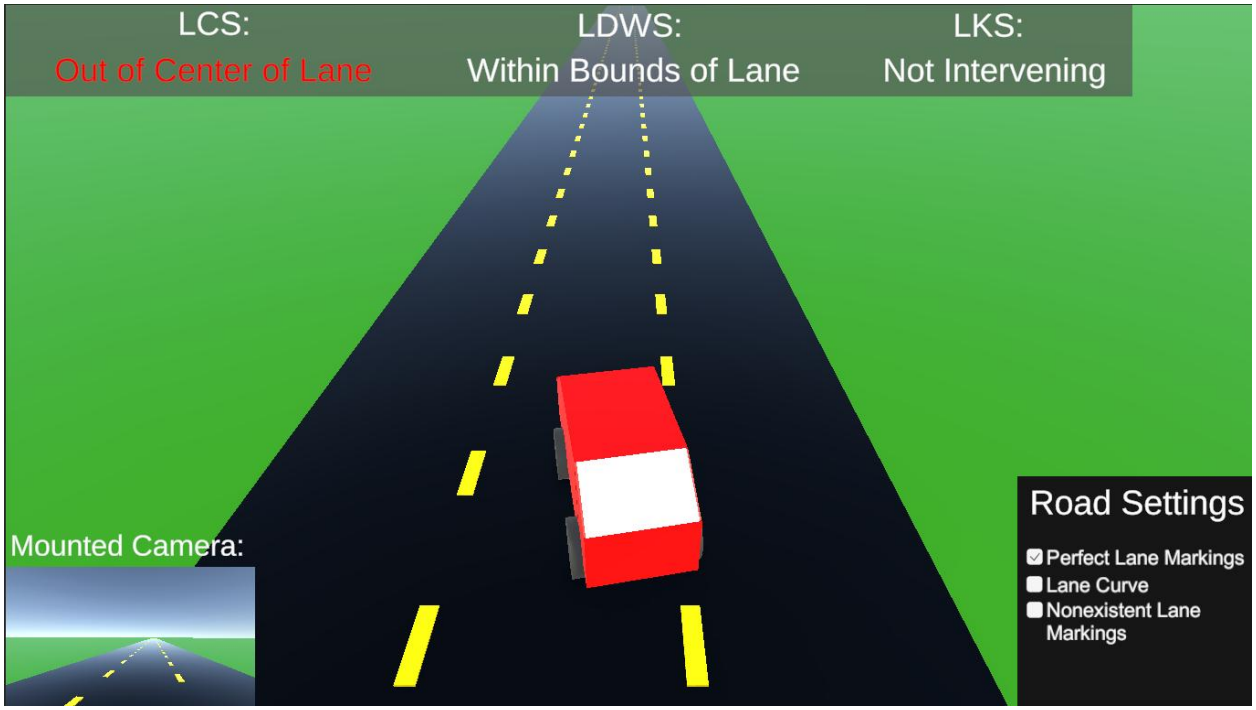


Figure 21: The driver partially merges back into the center lane.

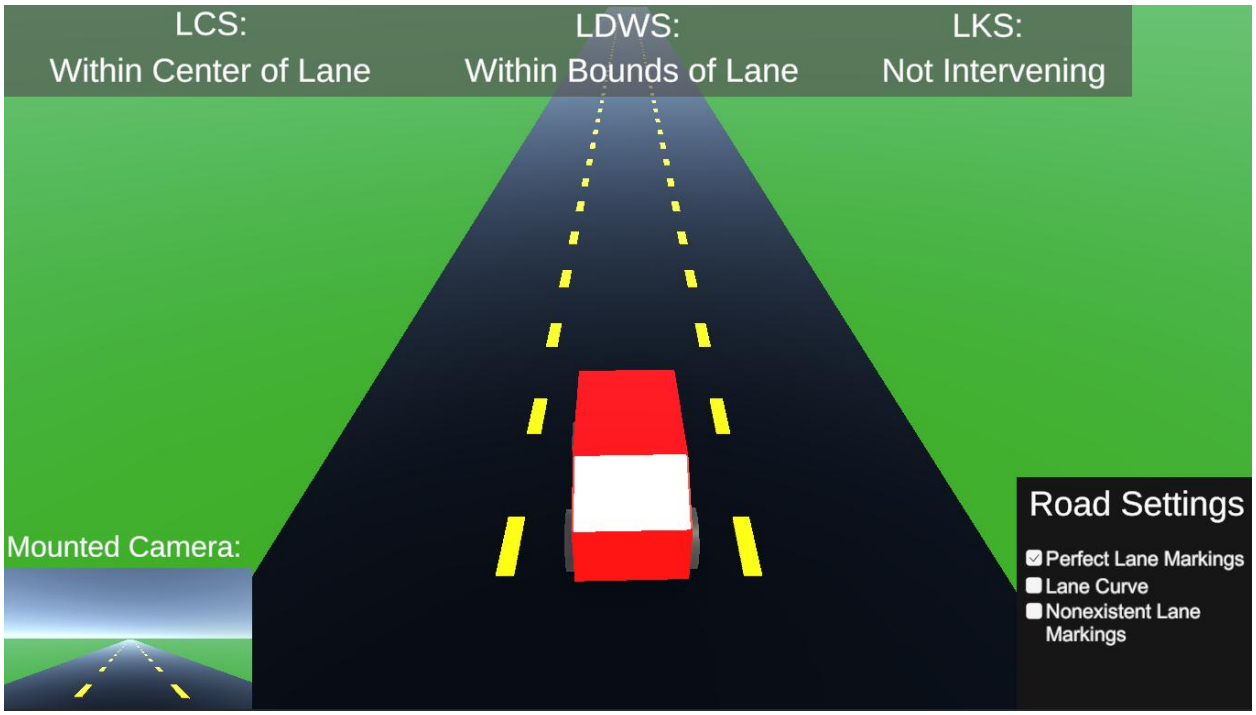


Figure 22: The driver fully merges back into the center lane.

6. References

[1] M. R. Pasha, S. Latha, R. Usha, V. B. Marin, M. V. Ishwarya, and M. Muthulekshmi, "Dynamic Lane Management with IoT for Real-time Lane Configuration and Traffic Flow," in Proc. 2024 2nd Int. Conf. on Computer, Communication and Control (IC4), Feb. 2024, pp. 1–5.

[2] Z. H. Khattak, B. Smith, M. Fontaine, J. Ma, and A. Khattak, "Active lane management and control using connected and automated vehicles in a mixed traffic environment," Transp. Res. Part C Emerg. Technol., vol. 139, 2022, Art. no. 103648.

[3] Chen, J. Team LMS2 Group Website. https://g3n3ric.github.io/. Accessed November 10th.

7. Point of Contact

For further information regarding this document and project, please contact **Prof. Betty H.C. Cheng** at Michigan State University (chengb at msu.edu). All materials in this document have been sanitized for proprietary data. The students and the instructor gratefully acknowledge the participation of our industrial collaborators.